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IRRIGATION AND WATER QUALITY

LAB REPORT 2

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MMAMASHIA WATER TREATMENT PROCESS

SAFETY

The Mmamashia Water Treatment Plant operates under strict safety protocols to ensure the protection of personnel from physical, chemical and operational hazards associated with water treatment processes. Personal Protective Equipment (PPE), including safety boots, gloves, protective clothing and goggles, is mandatory when working within the plant, particularly during chemical handling and maintenance operations.

Significant risks are associated with the chemical dosing stage, where coagulants and disinfectants are used. Chemicals such as polymer coagulants and chlorine-based disinfectants can be hazardous if inhaled or if they encounter the skin or eyes. Therefore, proper storage, accurate labeling, and controlled dosing systems are essential to minimize exposure and ensure safe handling. Mechanical hazards are present in equipment such as pumps, mixers, and filtration units. Regular inspection and maintenance of machinery are required to prevent equipment failure and accidents. Lockout and isolation procedures must be followed during repairs to prevent unintended activation of equipment.

Electrical safety is also critical due to the use of automated monitoring and control systems. All electrical installations must be properly insulated and clearly labelled to reduce the risk of electric shock. In addition, safety signage, restricted access areas, and emergency response procedures are implemented throughout the plant to enhance awareness and ensure a safe working environment.



The photo above shows some safety signs.

INTRODUCTION

The Mmamashia Water Treatment Plant plays a crucial role in supplying safe and reliable drinking water to Gaborone and its surrounding areas. According to reports from Mmegi Online (2014), the plant is one of the largest water treatment facilities in the region and plays a central role in supplying water. Water is an essential natural resource required for human survival, economic development, and environmental sustainability. However, the quality of water is equally important as its availability, since contaminated water can lead to serious public health issues such as waterborne diseases. Therefore, water treatment processes are necessary to ensure that raw water from natural sources is made safe for human consumption.

Water quality refers to the physical, chemical, and biological characteristics of water in relation to its suitability for a particular use, especially drinking. To maintain safe water standards, treatment plants are designed to remove impurities such as suspended solids, dissolved substances, and harmful microorganisms. In Botswana, the responsibility for water supply and treatment lies with the Water Utilities Corporation.

The plant treats water obtained from multiple sources, including dams and groundwater systems, which often vary in quality due to seasonal and environmental factors (WHO, 2017). As a result, effective treatment and continuous monitoring are required to ensure that the final water supplied meets the standards set by the World Health Organization (WHO) and Botswana Bureau of Standards and of recent Water Utilities Corporation has set their standards for the water quality.

In Botswana, water scarcity and increasing demand due to urbanization have made efficient water treatment systems essential. The plant forms part of the North-South Carrier system, which transports raw water from various dams and wellfields to the treatment facility. Its primary function is to remove physical, chemical and biological contaminants from raw water to make it suitable for human consumption. Given the variability in water quality from different sources, the plant is designed to handle a wide range of treatment challenges, ensuring consistent water quality.

AIMS

- To evaluate the water treatment processes at the Mmamashia Water Treatment Plant, with emphasis on process efficiency, water quality improvement, and operational performance.

OBJECTIVES

- To describe the sequence of treatment processes: screening, coagulation, flocculation, sedimentation, filtration, disinfection storage and distribution.
- To assess the effectiveness of each treatment stage in removing turbidity, suspended solids and microorganisms.
- To evaluate how the plant adapts to varying raw water quality from different sources.
- To examine the role of the plant in ensuring a reliable and safe water supply in Botswana.

METHODOLOGY

STUDY AREA/ DESCRIPTION OF THE AREA

The study was conducted at the Mmamashia Water Treatment Plant located approximately 15 km north of Gaborone. The plant is strategically positioned to supply potable water to the capital city and surrounding areas, making it a critical component of Botswana's water supply infrastructure.

The plant receives raw water from several major surface water sources including:

- Letsibogo Dam
- Dikgatlhong Dam
- Gaborone Dam
- Bokaa Dam
- Masama Wellfields

The region experiences a semi-arid climate characterized by low and unpredictable rainfall, high evaporation rates and frequent droughts. These conditions make water resource management particularly important. These sources differ in quality with surface water often containing high

turbidity and algal content, while groundwater may contain dissolved minerals. This variability necessitates a robust and adaptable treatment process.

DATA COLLECTION

Data was collected through:

- Direct observation of treatment processes
- Document Review

PIPELINE OF WATER

From the Letsibogo Dam the pipeline will run adjacent to the proposed Letsibogo Dam access road, passing round the village of Mmadinare to follow the existing road to Mmadinare from the Serule/Selebi-Phikwe road. It will then follow the BPC power line in a south-westerly direction bypassing Palapye to run parallel to the Gaborone-Francistown railway under the road to Martins Drift. It will then be located adjacent to the Gaborone-Francistown railway reserve as far as the Makoro Pump Station, before following the same railway to the Tewane Siding south of Radisele where it will pass just south of Lose Hill and cross the railway and main road before Lose Siding. From a point north of Lose Siding, it will be located adjacent to the Gaborone-Francistown main road reserve to a point north of Mahalapye and run across-country near the BPC power line to pass round the western outskirts of Mahalapye. It will then run adjacent to the Gaborone-Francistown main road reserve from a point south of Mahalapye to Serome Pump Station as far as a point north of Rasesa village and then continue cross-country west of the Gaborone-Francistown Road west of the Morawa Hills and close to the existing WUC Bokaa Dam and across to the inlet of the north Mmamashia treatment works.

WATER TREATMENT PROCESS

1. Water source – the Dam (Dikgatlhong and Bokaa)
2. Coagulation and Flocculation
3. Sedimentation
4. Filtration
5. Disinfection
6. Testing Lab
7. Storage
8. Distribution

Raw water is abstracted and conveyed from Dikgatlhong and Bokaa dam and gravity fed in the elevated tower where it will go in the plant with gravitational force. In water, the dirt in water is negatively charged. Negative particles repel each other, to allow the particles to come together a coagulant (Zetafloc 553L and Zetafloc A50) is added to raw water in a mixing chamber at the inlet the process is called Coagulation.

Zetafloc 553L and Zetafloc A50 are synthetic polymer coagulant aids used to enhance the efficiency of coagulation and flocculation processes at the Mmamashia Water Treatment Plant. These polymers function primarily through charge neutralization and polymer bridging, which are key mechanisms in particle aggregation. In water treatment, suspended particles are typically negatively charged and remain stable in solutions due to electrostatic repulsion. When polymers such as Zetafloc are added, they neutralize these charges and create bridges between particles, allowing them to bind together and form larger, heavier flocs.

Zetafloc 553L is more effective under conditions of low turbidity because it provides sufficient bridging without causing excessive floc density or chemical overdosing. In relatively clean water, fewer particles are present, so a lighter polymer ensures controlled floc formation and prevents unnecessary sludge production.

On the other hand, Zetafloc A50 is better suited for highly turbid water with a high concentration of suspended solids. The stronger polymer chains and higher charge density promote rapid aggregation of large numbers of particles, forming dense flocs that settle more efficiently during

sedimentation. This is particularly important when sludge production is high as it improves settling rates and reduces the load on filtration units.



The pictures above show the mixing tank which contains the Coagulants that are to be added in water to be able to bind particles together to form a sludge.

Unstable colloids will now attract each other to form micro flocs (Flocculation) which are slowly released to the sedimentation tanks/ clarifiers and the sedimentation process has a retention time of approximately two hours, which allows sufficient settling of particles and improves overall treatment efficiency.



The picture above shows a sedimentation tank.

The clarified water flows out from the top of the tanks, while the sludge is periodically removed. As this floc increases, they become denser, and they will be removed from the water by means of gravity or separation to form a sludge blanket. This process is called Sedimentation. The outlet of the sediment tank is at the top of the tank, and the clear water is removed by a surface channel called trough.

Clarified water and excess flocs are removed at the discharge pipes (De-sludging). From clarifiers it will then slowly go to sand filters where clarified water will be passed slowly through a layer of sand 60 to 120 cm thick to trap any suspended solids, and this process is known as filtration.

Water that has passed through the silica sand will go to the sump where it will be disinfected. A combination of hydrochloric acid and sodium chlorite to form Chlorine dioxide as the by-product is the most widely used disinfectant for the inactivation of waterborne pathogens in drinking water supplies.

A sample of clarified water from the sump is collected and taken to testing labs to test for turbidity together with free chlorine. The disinfected water goes to the master balancing reservoir (MBR) before it is distributed to the consumers. The master balancing reservoir is a bulk storage for water temporarily before distribution, and it stores a quantity of 80 megalitres (mgl) of water. This reservoir plays a key role in balancing supply and demand, maintaining system pressure, and providing emergency storage. From there, the treated water is pumped into the distribution network supplying Gaborone and surrounding areas.



The photo above shows a master balancing reservoir.

FILTRATION

During the filtration stage, digital monitoring systems are used to control and observe the performance of individual filters. An operator interface panel is installed to monitor parameters within the filtration units. These control panels display real time information such as filter status, flow rates, valve positions, and pressure conditions across the filter beds. They also indicate operational modes, including filtration and backwashing cycles.

The monitoring system allows operators to detect issues such as clogging, excessive head loss, or reduced filter efficiency. This automated and continuous monitoring improves operational efficiency, ensures consistent water quality, and reduces the risk of system failure. The use of such control systems highlights the integration of modern technology in water treatment processes to maintain optimal performance of filtration units.



The pictures above show the digital system used to monitor the filtration tanks.

FILTER BACKWASHING

Filter backwashing refers to pumping water backwards through the filters media, sometimes including intermittent use of compressed air during the process. Backwashing is a form of preventive maintenance so that the filter media can be reused.

WHY IS BACKWASHING DONE?

Backwashing is done to clean the filter media also known as silica sand and to prevent overflow, whereby water from the clarifiers cannot go to the filter tank because the silica sand is full and cannot process or filter any more water to the sump.

HOW BACKWASHING IS DONE.

- Close the inlet valve to stop water from getting into the filtration tank
- Open the outlet valve for water inside the filtration tank to get out
- Open the air scour and start the air blower for 5-10 minutes to provide aeration
- Stop the air blower after the required time and close the air scour
- Open the water pump scour and start to up-wash until the silica sand is seen
- Stop washing up and close the water pump scour
- Close outlet
- Open the inlet to let water into the filtration tank



The picture on the left shows a machine used to activate the backwash process in the filtration tank.

DESLUDGING

Desludging is the process of removing accumulated sludge (settled solid material) from the bottom of treatment units such as sedimentation tanks, clarifiers, or septic systems. In water treatment plants, sludge builds over time as suspended particles settle out during the clarification process. Regular desludging is necessary to maintain treatment efficiency, prevent loss of storage capacity, reduce the risk of blockages, and ensure good water quality. The removed sludge is then further treated or disposed of in an environmentally safe manner.

THE PROCESS OF DESLUDGING

During sedimentation, the heavier particles and flocs settle to the bottom of the tanks to form a sludge. The sedimentation tanks are designed with cone-shaped bottoms, and this conical design facilitates sludge collection and desludging. The settled material naturally slides down to the narrow tip of the cone, where it can be easily removed by mechanical scrapers or pumps (by suction). Efficient desludging prevents the accumulation of excess sludge and maintains the effective volume of the sedimentation tanks and ensures optimal removal of suspended solids. This design is particularly important when treating highly turbid water with high sludge production as it minimizes operational interruptions and improves overall plant efficiency.



After desludging, the sludge goes to the recovery tank and moves to the sludge tank. There is unthickened sludge which is a sludge that is combined with water; it will be air blown to avoid settling off the sludge at the bottom of the tank. The unthickened sludge goes to the thickener tank through the unthickened pumps. The thickener will separate the sludge from water, the sludge with less water is called the thickened sludge, and the water is called supernatant water. Therefore, the sludge will settle at the bottom and supernatant water will move through the supernatant pipes to the supernatant tank of which it will later be moved to the treatment plant, while the thickened sludge is moved to the drying beds and later used as manure.

SLUDGE TREATMENT PROCESS

There are processes step in sludge treatment process, and it is extremely important to properly treat such sludge to minimize its environmental repercussions.

SLUDGE THICKENING

The first step in the sewage sludge treatment plan is called thickening. In this step, the sewage sludge is thickened in a gravity thickener to reduce its overall volume, thus enabling the easy handling of the sludge. Hence, water in the sludge is removed in the thickening process.

SLUDGE DIGESTION

After amassing all the solids from the sewage sludge, the sludge digestion process begins. This is a biological process in which the organic solids present in the sludge are decomposed into stable substances. This process also helps reduce the total mass of solids, while destroying any present pathogens to enable easy dewatering. The dry solid sludge is heated and then sludge then flows into the second tank where it is converted by other bacteria to produce a mixture of carbon dioxide

(CO₂) and methane, after which the methane is collected and compressed before it goes to the storage tank.

DEWATERING

After retrieving useful gases and other byproducts, the remaining sludge is then dewatered before final disposal. Therefore, it is important to dry and water the sludge beforehand.

DISPOSAL

Once the sludge has been effectively dewatered, it is transported to the drying beds and left on the sun for fourteen (14) days and is then sold to the farmers so that it can be used as a fertilizer.

The plant incorporates a water recycling system in which water lost during various treatment processes, such as backwashing and desludging, is collected and stored in a designated tank. This water is then pumped back into the treatment process to minimize losses and improve overall efficiency. The reintroduced water is discharged into the system through an inlet structure, where it is mixed with incoming water to ensure uniform treatment.

LABORATORY

WATER QUALITY

It is a measure of the suitability of water relative to the requirements of any human need or purpose.

WATER QUALITY SAMPLING

Random sampling strategy is assumed that each sample has the same probability as other samples to be selected to serve as a representation of the entire water. Sampling containers are labeled.

The following are tested in a water sample.

- Turbidity
- Free Chlorine
- pH

- Electronic Conductivity

CHEMISTRY TESTS

TURBIDITY

Turbidity is the cloudiness or haziness of fluid caused by suspended solids that are usually invisible to the naked eye. The test is done every two hours daily. Also, turbid meter is used.

WHY DO A TURBIDITY TEST?

A turbid meter will measure the decrease in the passage of light through a water sample based on the amount of floating particles in the water. The WHO (World Health Organization) established that the turbidity of drinking water should not be more than 5NTU and should be ideally below 1NTU.

NTU = Nephelometric Turbidity Units



The photo above shows a multi-parameter.

HOW TURBIDITY IS TESTED IN A WATER SAMPLE.

- Make sure that the tubes are clean because dirty, scratched, or damaged tubes can cause inaccurate reading.
- Collect raw, clarified, and treated water in tubes and label each tube.

- Water in the tube should be below the meniscus.
- Close the tubes to prevent the entry of dust and dirt.
- Put the sample in a turbid meter and close it, the light inside the turbidity test will be able to identify the turbidity in the water
- Repeat the same process with the remaining tubes to identify turbidity

COLOR

Color in water is caused by dissolved organic matter and other substances. Treated water should be clear, with no noticeable color, as this affects consumer acceptability.

TASTE

AND

ODOR

These parameters are important for consumer satisfaction. Drinking water should be free from any unpleasant taste or odor, indicating the absence of contaminants.

pH

pH is a measure of the acidity or alkalinity of water. For drinking water, the acceptable pH range is typically between 6.5 and 8.5. Maintaining this range is essential for effective treatment processes and to prevent corrosion or scaling in the distribution system.

TOTAL

DISSOLVED

SOLIDS

(TDS)

TDS represents the concentration of dissolved substances in water, including minerals and salts. Levels below 1000 mg/L are generally acceptable, although values below 600 mg/L are preferred for better taste.

HARDNESS

It is mainly caused by calcium and magnesium ions. While there is no strict health limit, levels below 500 mg/L are recommended to prevent scaling and improve water usability.

ELECTRICAL

CONDUCTIVITY

Electrical conductivity measures the ability of water to conduct electricity, which is directly related to the concentration of dissolved ions such as salts and minerals. It is an important indicator of

water salinity and overall mineral content. High EC values may indicate elevated levels of dissolved solids, which can affect water taste and suitability for consumption. Typically, drinking water should have an EC value below approximately 1500 $\mu\text{S}/\text{cm}$, depending on guideline standards.

FREE

Free chlorine refers to the amount of chlorine available in water for disinfection after the treatment process. It is essential for killing harmful microorganisms and maintaining water safety during distribution. However, it must be carefully controlled to ensure effectiveness while avoiding health risks or unpleasant taste and odor. The recommended free chlorine residual in drinking water is typically between 0.2 and 0.5 mg/L.

CHLORINE

JAR TEST/FLOCCULATOR

The purpose of the laboratory jar test is to select and quantify a treatment program for removal of suspended solids or oil from raw water and a dilute process. Jar tests are conducted on a four or six-place gang stirrer which can be utilized to simulate mixing and settling conditions in a clarifier. A Coagulant is added and mixed with water to determine the water treatment plant coagulant dosage they can use based on the results carried out.



Containers are put under the stirrers so that the flocculator can stir raw water with the coagulant to determine the water treatment plant for coagulant dosage.

The laboratory also tests for;

- Chemical oxygen demand (COD)

CHEMICAL OXYGEN DEMAND

Chemical oxygen demand (COD) is the measure of the capacity of water to consume oxygen during the decomposition of organic matter in the water. In other words, it is the amount of oxygen that is needed to oxidize organic matter present in a quantity of water. It is an important parameter in water quality analysis as it helps to reduce risk to humans and the environment.

COD is a way of monitoring the efficiency of water treatment plants. If water is left untreated or partially treated, the discharged water contains effluent organics that can compete with downstream organisms for oxygen. COD has a key role in reducing the likelihood of pollutants causing any environmental damage.

IMPORTANCE OF CHEMICAL OXYGEN DEMAND

To prevent the wide range of health dangers to humans and the continued existence of some species, it is essential that water source quality is properly assessed before it is drawn off to water streams like Notwane River of which animals and some farmers use the water for irrigation. COD testing is a key part of this process.

COD can be used to:

- Analyze the effectiveness of wastewater treatment solutions
- Determine the effect of wastewater disposal on the environment.
- As an index for determining overall water quality.

High organic contamination in water discharged to tributaries and streams can have a range of impacts. These include:

- Toxicity of organic compounds: health effects on plants and wildlife
- Decreasing dissolved oxygen and eutrophication

- Impact on fish populations

COD measures the amount of oxygen necessary to break down the organic substances that are pollutants in water. A higher COD in a sample indicates that it contains higher levels of oxidizable material. If this is the case, then the water will have reduced dissolved oxygen levels. Where this happens, the effects can be environmentally damaging to higher aquatic life-forms. The aim of wastewater treatment is to reduce levels of COD in water.

SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

It is an advanced computer-based system used to monitor and control industrial processes in real time. At the Mmamashia Water Treatment Plant, SCADA is used to oversee various stages of the water treatment process, including filtration, chemical dosing, pumping, and storage.

The system collects data from sensors and instruments installed throughout the plant, such as flow meters, pressure gauges, and turbidity sensors. This data is transmitted to a central control system where operators can monitor plant performance through graphical interfaces, such as the digital panels used in the filtration section. SCADA allows operators to observe key parameters including flow rates, water levels, valve positions, and filter conditions in real time.

In addition to monitoring, SCADA enables control of equipment such as pumps, valves, and backwashing systems. It can automatically trigger alarms when abnormal conditions occur, such as high turbidity or excessive pressure loss in filters. This improves response time, reduces human error, and ensures continuous operation of the plant.



The picture above shows a SCADA

RECOMMENDATIONS

- Regular monitoring of water quality parameters such as turbidity, pH, and residual chlorine should be maintained to ensure compliance with standards.
- Chemical dosing (e.g., coagulants and polymers like Zetafloc 553L and Zetafloc A50) should be optimized based on raw water quality to improve efficiency and reduce costs.
- Filtration systems should be maintained regularly, including timely backwashing to prevent clogging and ensure effective operation.
- Sludge management practices should be improved to ensure proper handling, treatment, and safe disposal or reuse.
- Regular maintenance of equipment such as pumps, valves, and tanks should be carried out to prevent breakdowns.
- Public awareness on water conservation should be promoted to reduce demand on the treatment plant.

CONCLUSION

- The Mmamashia Water Treatment Plant plays a critical role in supplying safe drinking water to Gaborone and surrounding areas.
- The treatment processes, including coagulation, flocculation, sedimentation, filtration, and disinfection, effectively remove contaminants from raw water.
- Digital monitoring systems enhance control and performance of filtration processes.
- The cone-shaped sedimentation tanks improve desludging efficiency and overall plant operation.
- Despite challenges such as varying raw water quality, the plant consistently produces potable water.
- The plant remains a key infrastructure supporting public health and water security in Botswana.

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